

04 — Calculations & Data Rules

Deterministic rules first. AI must never replace these calculations.

Date & Cycle Data Model

- Store dates as date-only values where possible. Store timestamps separately for audit/reminder purposes.
- Cycle Day 1 = first day of menstrual bleeding as entered by the user.
- Cycle length = difference between the first day of consecutive periods.
- Keep all user-entered dates immutable in the audit history; corrections should create an updated record rather than silently changing history.

Cycle Estimates

- Initial MVP can estimate average cycle length from recent completed cycles.
- If insufficient history exists, allow the user to set an expected cycle length and clearly label the result as an estimate.
- Do not use a fixed 28-day assumption when user data exists.
- Prediction confidence should decrease when historical cycles are highly variable.

Ovulation/Fertile Estimate

- For a simple calendar model, estimated ovulation may be approximated as cycle length minus 14 days, with the fertile window treated as the 5 days before estimated ovulation plus the estimated ovulation day.
- This is an estimate, not a confirmed ovulation event. The UI must show an uncertainty notice.
- Later versions may combine user-entered LH test results, cervical mucus observations and basal-body-temperature data, but the combined algorithm must be medically reviewed before release.

Pregnancy Dating

- For MVP, calculate estimated due date using the user's last menstrual period date with a standard 280-day/40-week gestational framework, while clearly labeling it as an estimated due date.
- If the user has clinician-provided dating information, allow that information to override the app's simple LMP estimate, with a source field such as 'Clinician/ultrasound'.
- Gestational age should be derived from the selected dating method and current date, not independently guessed by AI.

Pregnancy Week Display

- Show completed weeks + days, e.g. '9 weeks 2 days'.
- Trimester boundaries must be defined in one centralized configuration, not duplicated across UI code.
- Due-date countdown must never show negative or impossible values. After the estimated due date, show 'Estimated due date passed' rather than negative days.

Testing Strategy

- Unit test leap years, month boundaries, year boundaries and timezone changes.
- Test short and long cycles, missing periods, corrected dates and irregular cycles.

- Test pregnancy dates across leap years.
- Create golden test cases for every calculation before building the UI around it.

Source note

- Medical content in this document is product guidance, not a diagnosis or treatment protocol. Key fertility guidance references: American Society for Reproductive Medicine (ASRM), Optimizing Natural Fertility (2022), and ASRM Definition of Infertility (2023). Pregnancy urgent-symptom concepts should be aligned with current local clinical guidance. Sources: <https://www.asrm.org/practice-guidance/practice-committee-documents/optimizing-natural-fertility-a-committee-opinion-2021/> ;
<https://www.asrm.org/practice-guidance/practice-committee-documents/definition-of-infertility/>